MSG No. 192.248 $P6'/m'c'c$ [Type III, hexagonal]

Table 1: Wyckoff site: 2a, site symmetry: 6'2'2

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	[1,2,3,4,5,6,13,14,15,16,17,18]
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	[7,8,9,10,11,12,19,20,21,22,23,24]

Table 2: Wyckoff site: 2b, site symmetry: 6'/m'..

No.	position	mapping
1	$[0,\ 0,\ 0]$	[1,2,3,7,8,9,13,14,15,19,20,21]
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	[4,5,6,10,11,12,16,17,18,22,23,24]

Table 3: Wyckoff site: 4c, site symmetry: 3.2

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	[1,2,3,4,5,6]
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	[7,8,9,10,11,12]
3	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	[13,14,15,16,17,18]
4	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	[19,20,21,22,23,24]

Table 4: Wyckoff site: 4d, site symmetry: -6'..

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	[1,2,3,19,20,21]
2	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	[4,5,6,22,23,24]
3	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	[7,8,9,13,14,15]
4	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	[10,11,12,16,17,18]

Table 5: Wyckoff site: 4e, site symmetry: 6'..

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,2,3,13,14,15]
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[4,5,6,16,17,18]
3	$[0,\ 0,\ -z]$	[7,8,9,19,20,21]
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[10,11,12,22,23,24]

Table 6: Wyckoff site: 6f, site symmetry: 2'2'2

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[1,5,14,16]
2	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	[2,6,15,17]
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[3,4,13,18]

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[7,11,20,22]
5	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[8,12,21,23]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[9,10,19,24]

Table 7: Wyckoff site: 6g, site symmetry: 2'/m'..

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,7,14,20]
2	$\left[0, \frac{1}{2}, 0\right]$	[2,8,15,21]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[3,9,13,19]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[4,10,18,24]
5	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[5,11,16,22]
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[6,12,17,23]

Table 8: Wyckoff site: 8h, site symmetry: 3..

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,2,3]
2	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	[4,5,6]
3	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[7,8,9]
4	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	[10,11,12]
5	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[13,14,15]
6	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2} - z\right]$	[16,17,18]
7	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[19,20,21]
8	$\left[\frac{1}{3}, \frac{2}{3}, z + \frac{1}{2}\right]$	[22,23,24]

Table 9: Wyckoff site: 12i, site symmetry: 2'..

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	[1,14]
2	$\left[0, \frac{1}{2}, z\right]$	[2,15]
3	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[3,13]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	[4,18]
5	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[5,16]
6	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[6,17]
7	$\left[\frac{1}{2}, 0, -z\right]$	[7,20]
8	$\left[0, \frac{1}{2}, -z\right]$	[8,21]
9	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[9,19]
10	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	[10,24]
11	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	[11,22]
12	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	[12,23]

Table 10: Wyckoff site: `12j`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,16]`
2	$\left[0,\ x,\ \frac{1}{4}\right]$	`[2,17]`
3	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[3,18]`
4	$\left[x,\ x,\ \frac{1}{4}\right]$	`[4,13]`
5	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[5,14]`
6	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[6,15]`
7	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[7,22]`
8	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[8,23]`
9	$\left[x,\ x,\ \frac{3}{4}\right]$	`[9,24]`
10	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[10,19]`
11	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[11,20]`
12	$\left[0,\ x,\ \frac{3}{4}\right]$	`[12,21]`

Table 11: Wyckoff site: `12k`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1,5]`
2	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[2,6]`
3	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[3,4]`
4	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[7,11]`
5	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[8,12]`
6	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[9,10]`
7	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[13,18]`
8	$\left[-x,\ -2x,\ \frac{1}{4}\right]$	`[14,16]`
9	$\left[2x,\ x,\ \frac{1}{4}\right]$	`[15,17]`
10	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[19,24]`
11	$\left[x,\ 2x,\ \frac{3}{4}\right]$	`[20,22]`
12	$\left[-2x,\ -x,\ \frac{3}{4}\right]$	`[21,23]`

Table 12: Wyckoff site: `12l`, site symmetry: `m'..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,20]`
2	$[-y,\ x-y,\ 0]$	`[2,21]`
3	$[-x+y,\ -x,\ 0]$	`[3,19]`
4	$\left[x,\ x-y,\ \frac{1}{2}\right]$	`[4,24]`
5	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	`[5,22]`
6	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[6,23]`
7	$[-x,\ -y,\ 0]$	`[7,14]`
8	$[y,\ -x+y,\ 0]$	`[8,15]`
9	$[x-y,\ x,\ 0]$	`[9,13]`
10	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	`[10,18]`
11	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	`[11,16]`
12	$\left[y,\ x,\ \frac{1}{2}\right]$	`[12,17]`

Table 13: Wyckoff site: `24m`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x-y,\ z]$	`[2]`
3	$[-x+y,\ -x,\ z]$	`[3]`
4	$\left[x,\ x-y,\ \frac{1}{2}-z\right]$	`[4]`
5	$\left[-x+y,\ y,\ \frac{1}{2}-z\right]$	`[5]`
6	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	`[6]`
7	$[-x,\ -y,\ -z]$	`[7]`
8	$[y,\ -x+y,\ -z]$	`[8]`
9	$[x-y,\ x,\ -z]$	`[9]`
10	$\left[-x,\ -x+y,\ z+\frac{1}{2}\right]$	`[10]`
11	$\left[x-y,\ -y,\ z+\frac{1}{2}\right]$	`[11]`
12	$\left[y,\ x,\ z+\frac{1}{2}\right]$	`[12]`
13	$[x-y,\ x,\ z]$	`[13]`
14	$[-x,\ -y,\ z]$	`[14]`
15	$[y,\ -x+y,\ z]$	`[15]`
16	$\left[x-y,\ -y,\ \frac{1}{2}-z\right]$	`[16]`
17	$\left[y,\ x,\ \frac{1}{2}-z\right]$	`[17]`
18	$\left[-x,\ -x+y,\ \frac{1}{2}-z\right]$	`[18]`
19	$[-x+y,\ -x,\ -z]$	`[19]`
20	$[x,\ y,\ -z]$	`[20]`
21	$[-y,\ x-y,\ -z]$	`[21]`
22	$\left[-x+y,\ y,\ z+\frac{1}{2}\right]$	`[22]`
23	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	`[23]`
24	$\left[x,\ x-y,\ z+\frac{1}{2}\right]$	`[24]`